

Module: EE6003 Embedded Systems

Semester: 2, 2025-26

Advanced Industrial Robotic Arm using RF Wireless Communication for Industrial Applications

Due: 25-June-2026

Submission details: Report submission in Canvas as a file upload

Assessment type: Coursework Group report

Assignment weighting: 100%

Module Learning Outcomes: 1, 2,3,4

KSBs: B2, B3, B4, B7, B12, K1, K2, K3, K4, K7, S2, S3, S4, S9

Task outline:

The aim of this coursework is to design and develop an embedded system-based industrial robotic arm controller that utilises Bluetooth or any wireless communication, closed-loop position feedback, and a dual embedded-unit architecture to deliver accurate, repeatable, and real-time operation.

The system consists of two units:

1. **Control Unit (Transmitter):** A handheld controller that must include a joystick or joint-mimic controller as the primary manual input. Processes raw physical movements into digital command packets sent via wireless communication to the receiver.
2. **Robotic Arm Unit (Receiver):** A robotic arm with a minimum of 3 degrees of freedom (3 DOF) plus the gripper. The arm shall perform both manual joystick or mimic version operation and self-autonomous operation:
 - **Manual operation (User-Led Control):** The operator uses handheld or joint-mimic controller to move the robotic arm in real-time. The system must be highly responsive, ensuring the arm moves smoothly and follows the operator's input without noticeable delay. Joints must use potentiometers (or similar sensors) to constantly track their actual position. This ensures the arm stays accurate and corrects itself if the motors shift or carry heavy loads.

- **System-Led Control (Smart Mode):** A smart mode where the robot utilizes "Sense-Think-Act" logic to perform tasks independently.
- The Sequence: Scan Area → Detect Object (Short-range IR) → Identify Colour → Move to Bin → Increment Counter.
 - The Goal: Automatically sort objects until a target count of a minimum of 5 items is reached.
 - Code Optimization: It is preferred that students implement Finite State Machines (FSMs) to ensure the firmware is optimized, predictable, and modular.

Note: Students are required to design a robotic arm with gripper using NXdesign (or any CAD software) and fabricate it through 3D printing or another suitable manufacturing method.

Deliverables:

Project Objectives

1. Develop an embedded system consisting of two units:
 - A control unit that captures joystick input or joint-mimic controller and transmits commands via Bluetooth. (Additional input devices (buttons/switches/potentiometers/encoders/custom commands) may be added for student creativity, (For example including support for ISR/RTOS features, Finite State Machine (FSM), safety actions, and command-based control))
 - A robotic arm unit that receives commands and performs accurate joint motion.
2. Implement Bluetooth or any suitable wireless communication for reliable and low-latency wireless control.
3. Ensure the robotic arm has a minimum of 3 DOF, capable of coordinated multi-joint movement.
4. Integrate potentiometers (or equivalent sensors) for closed-loop position feedback to verify and regulate actual joint positions.
5. Program motor control for:
 - Independent joint movement
 - Coordinated motion sequences
 - Gripper operation (open/close and, if attempted, variable control)
6. Implement two operating modes:
 - Manual mode (mandatory): Joystick-based real-time control
 - Self-autonomous mode (mandatory): Predefined or logic-driven sequence execution without continuous joystick input.
7. Ensure safe and stable mode switching between manual and autonomous control.
8. Optimise response time, stability, and power behaviour of the embedded system.
9. Evaluate system performance using accuracy, latency, repeatability, and robustness metrics.
10. Document the entire development process, including hardware schematics, firmware, CAD files, testing and analysis.

Assessment Structure and Individual Contribution

Each group will submit a group report, but students must clearly identify their individual contributions in the following areas:

1. Abstract, Introduction, Literature Review
2. Embedded System Design & Architecture
3. Hardware Development
4. Software Development
5. Robotic Arm Design & 3D Printing
6. Testing & Performance Evaluation
7. Teamwork And Professional Feedback
8. Report Writing & Documentation

Marks Distribution

Assessment Criteria	Marks
Abstract, Introduction, Literature Review	10
Embedded System Design & Architecture	10
Hardware Development & Sensor Integration	20
Software Development & Bluetooth Communication	15
Robotic Arm Design & 3D Printing	10
Testing & Performance Evaluation	20
Teamwork And Professional Feedback	10
Report Writing & Documentation	5

IMPORTANT NOTE ON CORE AND ADVANCED WEIGHTAGE (Must Appear in Report)

- Core Coursework Weightage: 60%
- Advanced Embedded Systems Weightage: 40%

Students who correctly implement all core coursework requirements may achieve marks up to a maximum of 60% (upper 2:1 threshold). Marks above 60% will be awarded only where students demonstrate advanced or novel embedded systems features that clearly reflect embedded systems concepts and principles, such as real-time behaviour (FreeRTOS), interrupt-driven design, State machines, closed-loop control optimisation, communication protocol design, robustness, safety modes, and power management etc.

Students must explicitly identify and discuss:

- Which elements of their work address core requirements, and
- Which elements represent advanced embedded systems features

within the relevant sections of the technical report.

Submission Guidelines:

- **Report:** A well-structured Individual academic report (4500 words) in IEEE Letter format.
- **A zip file containing all supplementary materials** (for example: code, videos, or sensor readings referenced in the report)
- Clearly specified individual contributions from each group member

Plagiarism Policy: Any form of plagiarism will result in disqualification.

Late submission: You will be capped to 40% for the first 5 days and 0% thereafter unless you have extenuating circumstances

Team Composition:

The coursework will be a combination of group work, with an overall design being the responsibility of the group and an individual design element or contribution to be included as part of the final submission. Each team will consist of 3 members. The organization of the team will be entirely up to the team members.

Guidance for Technical Report:

This guidance provides and outline of the key aspects to include in the technical report, but you are encouraged to enhance the report by adding your own insights explanations, justification of the choices made and analysis. A well-structured and critically examined will be assessed accordingly.

1. Abstract, Background, Literature Review (10 Marks)

- Industrial automation needs, role of multi-DOF robotic arms with grippers in modern manufacturing.
- Overview of Bluetooth and alternative protocols (e.g., Zigbee), latency and interference issues.
- Principles of feedback control, comparison of potentiometer, encoder, and force sensors.
- Analysis of existing systems, gaps in reliability, control precision, real-world constraints, and innovation opportunities.
- Evidence your understanding of why reliable, smart wireless control and feedback are crucial for industrial pick-and-place.

2. Embedded System Design and Architecture (10 Marks)

- Comprehensive Block Diagram: Clearly show all major system modules including control input (joystick/gesture), wireless communication, arm joint actuation, and gripper with feedback sensors.
- Hardware Selection Justification: Compare microcontroller candidates, justify the chosen model for multitasking, wireless integration, and I/O capabilities.
- Wireless Communication: Explain both the Bluetooth module and any advanced dual-protocol solution. Clearly justify your choice in relation to industrial interference/security.
- Power Requirements & Management: Assess all unit consumption; discuss efficiency strategies (sleep modes, dynamic clocking).
- Mode Control Flow: Describe the logic (including state diagrams if relevant) for transitioning between manual and autonomous modes, especially how gripper actions are coordinated.

3. Hardware Implementation (20 Marks)

- Assembly of robotic arm and controller components (servo motors, sensors, etc.). List the sensors used for gesture detection (e.g., Joystick, flex sensors, IMU, EMG). Explain their working principles. How did you interface sensors with the microcontroller to read data accurately?

- Schematic diagram and describe the **circuit connections** for the transmitter and receiver units with a circuit diagram.
- Power management and circuit protection measures. Circuit protection, grounding, interference awareness. Hardware challenges and how they were resolved.

Note: To evaluate the Hardware Development and integration criterion in the marking rubric, the working prototype will be physically inspected. Students are therefore encouraged to demonstrate their prototype. In addition, a demonstration video must be submitted to support review and moderation. Each student should answer specific questions based on their contribution.

4. Software Development (15 Marks)

- Detailed description and flowchart/state diagram of the main control loop, including gripper actuation and feedback synchronization.
- Joystick Mapping & User Input Strategy
- Wireless Data Protocol: Specify packet format, payload structure, error checking, and methods you used to minimize latency (e.g., buffer size, priority handling, ISR).
- Operational Modes: Document manual mode (real-time control) and autonomous mode (predefined pick-and-place sequence using state machine).
- Feedback Loops: Explain how the software compares target vs. real positions, providing error correction and diagnostics.
- Documentation: Commented code sections, architecture diagrams, and design rationale for maintainability and scalability

5. Robotic Arm Design & 3D Printing (10 Marks)

- Provide a SolidWorks (or equivalent) design file of the robotic arm. Include screenshots.
- Describe the 3D printing or fabrication method used. Justify your choice of materials. Evidence of improvements.

6. Performance Evaluation and Testing (20 Marks)

- Describe the testing procedures used to verify correct finger movement such as functional testing of robotic arm movements.
- What metrics were used to measure communication accuracy and response time and how did you evaluate the power efficiency and battery life of the system? Discussion of system limitations and improvements
- Conclusion: Summarize your findings and key outcomes of the project. Clearly state possible improvements and realistic directions for future development based on technical challenges encountered.

7. Teamwork And Professional Feedback [10 marks]

As you complete this project you are expected to support your group members. Early on in this project you should identify the strengths of your team and allocate roles appropriately. You should also keep a record of the tasks that are agreed upon and the challenges you face as a team.

1. Provide professional, balanced, and constructive feedback for each of your group members. This should be submitted independently to the module lead via canvas.

You should consider:

- What were the primary contributions of the individual you are evaluating (technical work, leadership, support)?
- Why were their contributions impactful to the team?
- How could they have had more impact to the project?

3. Based on the feedback received from your peers your independent contributions will be assessed, and marks will be allocated for relative contribution, team synergy, and supportive practice.

8. Report Writing and Documentation (5 Marks)

- Proper citation and references: Ensure all statements, technical claims, design decisions, and literature reviews are backed by relevant published sources.
- Each section must include proper IEEE-style citations and a comprehensive reference list.
- Use clear, concise language and structure your report with well-defined headings and subheadings for logical progression.
- Support your written report with screenshots, diagrams, and a video demonstration that effectively showcases system functionality and validates your testing and results.

Compile all the above sections into a well-structured technical report. Ensure it includes the title and acknowledge individual contributions in the group.

Word count: Report: A well-structured academic report (4500 words) in IEEE Letter format. Any submission exceeding the stated assessment volume by more than 10% will have their grade capped at a passing grade (40%). Exceptions may be made to this in extenuating circumstances. (excluding appendices, code, figures, references, and tables).

Marking criteria:

Your work will be assessed against the following marking criteria.

	Fail (0-39%)	3 rd (40-49%)	2:2 (50-59%)	2:1 (60-69%)	1 st (70-79%)	1 st + (80-100%)
Abstract, Introduction Literature Review Learning Outcome: LO1, LO2, LO3, LO4	Minimal or irrelevant discussion, lacks citations, weak understanding of embedded systems and Bluetooth communication.	Basic discussion, some relevant literature but lacks depth and clarity.	Adequate research, but missing technical depth or critical analysis	Well-researched with clear understanding, supported by relevant literature.	Comprehensive analysis, well-structured, strong references, and technical insight.	Exceptional review with critical analysis, deep insights, and strong academic references
Embedded System Design & Architecture Learning Outcome: LO1	Poorly structured or missing diagrams, no clear explanation of system components.	Basic block diagram with limited details, unclear explanation of control strategies.	Functional design but lacks detail in sensor-actuator integration and communication protocols.	Well-structured system design, clear block diagrams, and justified architecture.	Highly detailed system architecture explains communication protocols (UART, Bluetooth etc) and control strategies effectively.	Exceptional system design, clear justifications, well-optimized architecture, strong technical depth.
Hardware Development & Sensor Integration LO2, LO3, LO4	System does not function or lacks key components (e.g., missing Bluetooth communication or motor control).	Basic system with partial functionality but has major issues in responsiveness or accuracy.	Functional system but noticeable delays, inefficiencies, or incomplete gesture recognition	Well-implemented system with effective Bluetooth control, real-time processing, and Fully functional, safe, and correctly interfaced hardware meeting coursework requirements.	Highly optimized system with smooth operation, real-time performance, and seamless Bluetooth control. Hardware enhanced through feedback optimisation, noise handling, or robustness improvements.	Outstanding implementation with excellent real-time synchronization, efficiency, and robustness. Industrial-quality hardware design with strong reliability, protection, and performance optimisation.
Software Development & Bluetooth Communication Learning Outcome: LO2, LO3, LO4	Code is incomplete, disorganized, or non-functional.	Basic embedded C/C++ code /python with multiple inefficiencies and missing modularity.	Functional firmware with some response delays or lack of efficiency. Functional firmware but limited modularity or optimisation	Well-structured code with efficient sensor-actuator synchronization and Bluetooth communication. Well-structured firmware supporting manual and autonomous modes reliably.	optimized embedded system firmware efficient and well-commented. Advanced software design using ISRs, FreeRTOS, state machines, or structured protocols	Highly efficient, modular and well documented code with real-time performance optimizations , robust error handling. Highly optimised real-time firmware using RTOS, deterministic timing, and robust error handling.

Robotic Arm Design & 3D Printing Learning Outcome: LO2, LO3, LO4	Poor or missing CAD model, no fabrication attempt.	Basic CAD model with significant design flaws, fabrication may be incomplete.	Functional CAD model with minor flaws, partially fabricated.	Well-designed robotic Arm, successfully fabricated but with some minor improvements needed.	Highly detailed CAD model, well-printed and fully functional prosthetic hand.	Exceptional CAD design, optimized for stability, usability, and real-world application.
Testing & Performance Evaluation Learning Outcome: LO2, LO3, LO4	No testing conducted, no evaluation of accuracy or performance.	Minimal testing with weak analysis of system performance. Basic functional testing only.	Basic validation of functionality, but lacks detailed performance evaluation. Structured testing with evaluation of accuracy, latency, and repeatability	Well-structured testing methodology, evaluation of accuracy, latency, and power consumption.	Comprehensive testing, detailed analysis, and clear discussion of system performance improvements. Testing used to guide optimisation and system refinement	Exceptional evaluation with advanced optimization techniques and in-depth technical analysis. Extensive quantitative testing with benchmarking and deep critical analysis.
Teamwork And Professional Feedback	Poor collaboration, lack of contribution from team members	Minimal Team work, work distribution unclear and does not add any va	Adequate team work but some member contributed more than others	Good collaboration, work well distributed among team members Clear roles, effective collaboration, professional conduct	Strong teamwork with clear roles and good communication and innovative contributions enhancing project outcomes. Strong teamwork enhancing technical outcomes and innovation	Exceptional team work with outstanding coordination and equal contribution and significant enhancements to project outcome though innovative ideas Exceptional leadership, coordination, and collective technical excellence.
Report writing and Documentation	Poorly structured report with significant missing technical details, lacks coherence.	Basic report with weak explanations and minimal embedded systems terminology.	Adequate report but lacks technical depth or clarity in some sections with proper citations	Well-structured technical report with clear explanations, good use of embedded system concepts.	Highly detailed report, well-organized, strong technical depth, and proper citations.	Exceptionally written technical report with professional formatting, critical analysis, and advanced insights.

Note: To evaluate the Hardware Development and integration criterion in the marking rubric, the working prototype will be physically inspected. Students are therefore encouraged to demonstrate their prototype. In addition, a demonstration video must be submitted to support review and moderation. Each student should answer specific questions based on their contribution.

<Statement of how group marks will be adjusted to give individual marks, where appropriate>

Feedforward opportunities:

Statement of the support available to students in the preparation of their work for assessment.

Feedback details:

The individual and collective written feedback and grades will be available through Canvas three weeks after the submission deadline.

If you have any queries, please contact the Module lead Srilatha Pamuri, and arrange a session for additional feedback.

Assessment Appeals:

If you have queries about your grade, please speak with your module lead for clarification. If the issue is not resolved, you may submit a formal appeal or raise a concern through the portal. The Governance Team will then review your case and take appropriate action.

Appeals should include any supporting evidence and be submitted within 10 working days of receiving the grade. For full details on the appeals process and permissible grounds, please read the Academic Appeals Policy on the website or the policy page on the Hub.

If you have any questions or concerns, please contact the Governance Team at studentcases@dyson.com.

Mitigating Circumstances:

If due to exceptional circumstances you are unable to submit the coursework at the set deadline, despite reasonable adjustments (if any), you are requested to apply for **Type 1** Mitigating Circumstances. Please ensure that you have consulted either the Module Lead, a member of the Student Support Advisor, or the Safeguarding team before filling in the application. The form, found on your OnTrack Learner Hub, should be submitted within the specified timeframes in the policy, otherwise, it will be deemed invalid.

The latest Mitigating Circumstances policy can be found here: [Mitigating Circumstances Policy 2025-2026](#)

Academic Misconduct:

By submitting coursework, electronically and/or hard copy, you state that you fully understand and are complying with the Institute's policy on Academic Integrity and Academic Misconduct. The policy can be found under: [Academic Integrity and Misconduct Policy 2025-26](#)

The consequences of committing academic misconduct can be extremely serious and may have a profound effect on your results and/or future progression. The penalties range from a written reprimand to cancellation of all of your marks and withdrawal from the Institute.

NB: You are reminded that in submitting any coursework for assessment you are confirming it is your own work as per the information laid out in the Academic Regulations and Academic Misconduct policies. If you have any questions, or require further clarification please contact the Module Leader.